Generated report 1: pbmc

Objective (as given)

"Learn a low-dimensional representation and generate realistic synthetic cells for sharing"

Agent's decision

Task: representation_learning | decision: proceed | task run: representation_learning | also: data_generation

profile.json reports 2,700 cells, well above the 200-sample floor, and neither task needs labels (has_labels is false, which is fine here). The data are raw non-negative integer UMI counts (is_nonneg_integer true) with 97.4% sparsity over 32,738 genes, which is the setting count-likelihood latent-variable models (e.g. a negative-binomial VAE) are designed for: one model gives both a latent representation and a sampler. A simpler non-generative method (PCA on log-normalized counts) could give a representation, but it cannot generate count-valued cells, so the report should compare the learned latent space against a PCA reference rather than treat the generative model as the only option. With only 2,700 cells, memorization is a real risk (DATA_DESCRIPTION.md says so), so held-out comparison is required.

Models run (headline metrics; details on the metrics page)

vae_nb_s0	vae	C2ST 0.9457 (real-vs-real 0.5855)	precision 0.7926	recall 0.7667	DCR ratio 1.4897
vae_nb_s1	vae	C2ST 0.9253 (real-vs-real 0.5664)	precision 0.8722	recall 0.7667	DCR ratio 1.4131
flow_pca_s0	flow	C2ST 0.8572 (real-vs-real 0.5855)	precision 0.5611	recall 0.987	DCR ratio 1.5933
flow_pca_s1	flow	C2ST 0.8352 (real-vs-real 0.5664)	precision 0.5315	recall 0.9833	DCR ratio 1.5531

Downstream-use verdicts

- [conditional] VAE latent codes as input to clustering or neighbor graphs (vae_nb_s0)

- [not_recommended] VAE synthetic cells in place of real cells (incl. sharing) (vae_nb_s0)

- [conditional] VAE synthetic cells for per-gene count summaries only (vae_nb_s0)

- [not_recommended] Flow synthetic cells in place of real cells (flow_pca_s1)

Summary (agent-written)

The agent trained a negative-binomial VAE on raw counts (2,000 genes) and a RealNVP flow on 20 PCA components, two seeds each. The VAE latent space preserves neighborhoods slightly worse than a same-size PCA (trustworthiness 0.8073 vs 0.8328). Synthetic cells match per-gene means and sparsity, but a classifier separates them from held-out real cells easily (VAE C2ST AUC 0.9457, flow 0.8572, real-vs-real reference 0.5855). The synthetic cells are not a substitute for real cells, and there is no formal privacy guarantee.

Objective interpretation

Interpretation

Primary task: representation_learning

Secondary tasks: data_generation

The objective asks for two things joined by 'and': 'learn a low-dimensional representation' maps to representation_learning, and 'generate realistic synthetic cells for sharing' maps to data_generation. Representation learning is listed first and is the primary task; it is also the part the evaluation can test most directly on 2,700 unlabeled cells. Generation is a secondary task evaluated with fidelity (C2ST, marginal statistics) and memorization (nearest-neighbor distance) checks against held-out real cells.

Success criterion

On held-out real cells, the model's reconstruction likelihood is close to its training value and the latent space preserves nearest-neighbor structure (trustworthiness) of the log-normalized data; and a classifier cannot reliably tell synthetic from held-out real cells (C2ST AUC close to 0.5), with synthetic cells no closer to training cells than held-out real cells are.

Whether to model, and how

Decision: proceed (task actually run: representation_learning)

profile.json reports 2,700 cells, well above the 200-sample floor, and neither task needs labels (has_labels is false, which is fine here). The data are raw non-negative integer UMI counts (is_nonneg_integer true) with 97.4% sparsity over 32,738 genes, which is the setting count-likelihood latent-variable models (e.g. a negative-binomial VAE) are designed for: one model gives both a latent representation and a sampler. A simpler non-generative method (PCA on log-normalized counts) could give a representation, but it cannot generate count-valued cells, so the report should compare the learned latent space against a PCA reference rather than treat the generative model as the only option. With only 2,700 cells, memorization is a real risk (DATA_DESCRIPTION.md says so), so held-out comparison is required.

Other readings considered

- 'For sharing' could be read as a request for privacy-preserving or anonymized data. This pipeline only checks memorization (synthetic-to-training nearest-neighbor distances compared with held-out-to-training distances); that is evidence, not a formal privacy guarantee such as differential privacy, and the output will not be described as private, anonymized, or safe to share.

- data_generation could have been chosen as the primary task since sharing synthetic cells is the stated end use. Representation learning was kept primary because it is named first and because the same latent-variable model serves both; generation quality is still evaluated and reported in full.

- 'Realistic' is not defined by the user. It is read as: synthetic cells match held-out real cells in per-gene mean/variance, sparsity, library size, and are hard to separate with a classifier. Biological realism (correct cell-type proportions, marker co-expression) cannot be checked directly because the dataset has no cell-type labels.

Data and preprocessing

Dataset (from DATA_DESCRIPTION.md)

Peripheral blood mononuclear cells (PBMCs) from a healthy donor, profiled with 10x Genomics single-cell RNA sequencing (scRNA-seq). This is the "pbmc3k" dataset used throughout the Scanpy tutorials and widely used as a benchmark for single-cell analysis methods. Stored as the raw h5ad file from the Scanpy tutorials.

Profile

2,700 samples x 32,738 features; modality tabular

Sparsity 0.9741; value range [0, 419]; non-negative integers: True

Labels: none

Held-out split

test_fraction 0.2, stratified: False -> 2160 training / 540 held-out samples

Reason: With 2,700 cells, a 0.2 fraction holds out 540 real cells for the C2ST, memorization, and held-out likelihood checks while leaving 2,160 for training (1,944 after the 10% validation slice train.py takes). stratify is false because profile.json reports has_labels false, so there is no class mix to preserve.

Preprocessing (applied identically for every model; fit on training data only)

- select_top_variance {"n_features": 2000}: profile.json reports 32,738 genes with 97.4% zeros; most genes are detected in almost no cells and would add noise and training time to a 2,160-cell model. Keeping the 2,000 most variable genes (ranked on normalized, log-transformed training cells only) keeps the genes that carry cell-to-cell differences. 2,000 is a judgment call: 1,000 would train faster but risk dropping weaker cell-type markers, 5,000 would mostly add near-zero genes.

- normalize_total : DATA_DESCRIPTION.md says library size varies about 10x between cells, so raw counts mix sequencing depth with biology. Normalizing to a common total defines the transformed view and the shared evaluation space, so distances between real and synthetic cells reflect expression profiles rather than depth. The NB-VAE still trains on raw counts with library size handled in its decoder.

- log1p : Counts are heavy-tailed (maximum 419 against a mean of 0.072 in profile.json), so a few highly expressed genes would dominate PCA and the classifier in the evaluation space. log1p compresses that range while keeping zeros at zero, which matters for a matrix that is 97.4% zeros.

Evaluation space

All models are scored in one shared space: PCA of the transformed data with 14 components (0.2258 of training variance), fit on the training split. The number of components is the largest k where an unseen half of the training data keeps >= 0.85 of the fitted half's cumulative PCA variance: a PCA with more components overfits the training data, and a classifier could then tell training from held-out real samples apart, biasing every comparison.

Models, hyperparameters, and training

vae_nb

Family: vae; input: raw; seeds [0, 1]; serves representation_learning, data_generation; status planned; estimated 1.5 min/seed

Why this model: Primary model: the objective asks for both a low-dimensional representation and synthetic cells, and a VAE is the only family here that gives a latent representation and a sampler from one fit. The NB likelihood on raw counts matches the integer, overdispersed, sparse data in profile.json. Two seeds because the sharing use depends on whether fidelity and memorization results are stable, and the budget (20 min) easily allows it. Its latent space is compared against PCA of the log-normalized data via trustworthiness, and its samples against the flow comparator and the real-vs-real reference.

- latent_dim = 10: n_train is about 2,160 cells, under 5,000; 10 latent dimensions is enough to capture the handful of major PBMC populations expected in this sample and keeps the representation small enough to inspect, while more dimensions would mostly fit noise with so few cells. Judgment call: 16 was the alternative.

- hidden_dims = [256, 128]: Input is 2,000 selected genes (at least 1,000 features), so a [256, 128] encoder compresses 2,000 to 10 dimensions in two steps without being large enough to memorize 1,944 training cells easily.

- epochs = 400: A maximum, not a target: early stopping on the 10% validation slice (patience 25) keeps the best weights. 400 covers the slow-converging case on 1,944 cells; a previous PBMC run stopped near epoch 91, so the cost is bounded.

- batch_size = 128: 128 gives about 15 gradient steps per epoch on 1,944 training cells, enough steps per epoch for stable Adam updates without tiny noisy batches.

- lr = 0.001: 1e-3 with Adam on a model this small (about 0.6M parameters for 2,000 inputs); lower only if the loss curve oscillates, which the run figures will show.

- likelihood = nb: profile.json reports is_nonneg_integer true and 97.4% sparsity: raw UMI counts are overdispersed, and DATA_DESCRIPTION.md names negative binomial as the usual choice. An NB decoder on raw counts generates integer-valued cells directly, which is what 'realistic synthetic cells' needs.

- kl_warmup_epochs = 40: About 10% of the 400-epoch maximum. Ramping the KL weight avoids the latent collapsing to the prior early, which would make the representation useless for the primary task.

Training (vae_nb_s0): 91 of 400 max epochs, kept epoch 66 (validation loss did not improve for 25 epochs); 0.194 min on cpu; 1944 fit / 216 validation samples

Training (vae_nb_s1): 98 of 400 max epochs, kept epoch 73 (validation loss did not improve for 25 epochs); 0.194 min on cpu; 1944 fit / 216 validation samples

flow_pca

Family: flow; input: pca; seeds [0, 1]; serves data_generation; status planned; estimated 0.5 min/seed

Why this model: Comparator for the secondary data_generation task: an exact-likelihood model

Models, hyperparameters, and training (cont.)

with a different inductive bias (continuous density on PCA of log-normalized counts, not a count decoder), scored in the same evaluation space as the VAE. If it matches or beats the VAE on C2ST, that tells us the VAE's count decoder is not the reason for good fidelity. It has no latent representation, so it does not serve the primary task. It produces continuous log-scale values, not integer counts, which limits what its samples can be used for.

- pca_components = 20: Density estimation needs many samples per dimension; with about 1,944 training cells, 20 components gives roughly 100 cells per dimension. The flow models these top 20 components of the log-normalized data and treats the remaining variance as independent Gaussian noise when generating, so its samples are full-dimensional but only the top components are learned. The retained variance is recorded in the run sidecar.

- n_layers = 8: 8 affine coupling layers is enough flexibility for a 20-dimensional, multi-population density without many parameters relative to 1,944 cells.

- hidden_dim = 128: 128 because n_train is about 2,000, far below the 20,000 where a wider network would be justified.

- epochs = 400: A maximum with early stopping (patience 25); on 20-dimensional input from these 1,944 cells each epoch is fast, so the cap costs little.

- batch_size = 128: 128 gives about 15 steps per epoch on 1,944 training cells, matching the VAE so the two are trained under comparable optimization settings.

- lr = 0.001: 1e-3 with Adam for a small coupling network on 20 inputs; the loss curve figure will show whether it needs lowering.

Training (flow_pca_s0): 38 of 400 max epochs, kept epoch 13 (validation loss did not improve for 25 epochs); 0.023 min on cpu; 1944 fit / 216 validation samples; input PCA keeps 0.2396 of variance

Training (flow_pca_s1): 39 of 400 max epochs, kept epoch 14 (validation loss did not improve for 25 epochs); 0.024 min on cpu; 1944 fit / 216 validation samples; input PCA keeps 0.2396 of variance

Evaluation focus

Primary: held-out vs training reconstruction likelihood and latent trustworthiness for the VAE, compared with PCA of the same data. Secondary: C2ST AUC between synthetic and held-out real cells, marginal statistics (per-gene mean/variance, sparsity), and synthetic-to-train nearest-neighbor distances compared with held-out-to-train distances, each next to its real-vs-real reference. Memorization checks are evidence only, not a privacy guarantee.

Evaluation metrics (held-out real data)

Fidelity

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
n_compared	540	540	540	540
mmd2	0.01961	0.01639	0.00767	0.00959
mmd2_reference	0.00026	-0.00028	0.00026	-0.00028
c2st_auc	0.9457	0.9253	0.8572	0.8352
c2st_auc_reference	0.5855	0.5664	0.5855	0.5664
precision	0.7926	0.8722	0.5611	0.5315
recall	0.7667	0.7667	0.987	0.9833

Memorization

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
dcr_median_synthetic	3.0228	2.8673	3.2329	3.1514
dcr_median_heldout	2.029	2.029	2.029	2.029
dcr_ratio	1.4897	1.4131	1.5933	1.5531
too_close_fraction	0.0056	0.0019	0	0.0074

Evidence about copying of training samples, not a formal privacy guarantee.

Feature stats

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
mean_corr	0.9978	0.9976	0.998	0.9976
var_corr	0.8745	0.8912	0.9694	0.9718
zero_frac_corr	0.9919	0.9916	n/a	n/a
zero_frac_synthetic	0.735	0.7357	n/a	n/a
zero_frac_heldout	0.7232	0.7232	n/a	n/a
library_median_ratio	0.9984	0.9896	n/a	n/a

Likelihood

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
train_mean	-1563.406	-1561.089	-25.368	-25.274
heldout_mean	-1586.174	-1585.719	-24.616	-24.54
per_dimension_heldout	-0.7931	-0.7929	-1.2308	-1.227

Likelihoods are in each model's own input space and units; do not compare them across models with different input spaces.

Representation

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
latent_dim	10	10	n/a	n/a
trustworthiness	0.8073	0.7917	n/a	n/a
trustworthiness_pca_baseline	0.8328	0.8327	n/a	n/a
knn_overlap	0.16	0.1475	n/a	n/a
knn_overlap_pca_baseline	0.1681	0.1677	n/a	n/a

Neighborhoods are Euclidean in the transformed space, which favors PCA (it is the best linear Euclidean summary by construction); a latent close to the baseline is not a failure.

How to read the metrics

- mmd2: Squared maximum mean discrepancy (RBF kernel, bandwidth = median distance among held-out real samples) between synthetic and held-out real samples in the evaluation PCA space. 0 = same distribution; compare with mmd2_reference.

- mmd2_reference: Same MMD, but a training-set subsample vs held-out real: the value two real samples of this size give.

- c2st_auc: Classifier two-sample test: cross-validated ROC AUC of a gradient-boosted classifier separating synthetic from held-out real samples. 0.5 = indistinguishable, 1.0 = perfectly separable.

- c2st_auc_reference: Same test, training subsample vs held-out real (should be near 0.5).

- precision: Fraction of synthetic samples inside the 5-NN radius of some held-out real sample (fidelity: are samples realistic?).

- recall: Fraction of held-out real samples inside the 5-NN radius of some synthetic sample (coverage: is the data's variety reproduced?).

- dcr_median_synthetic: Median distance from each synthetic sample to its closest training sample (evaluation space).

- dcr_median_heldout: Median distance from each held-out real sample to its closest training sample: the distance a genuinely new sample has.

- dcr_ratio: dcr_median_synthetic / dcr_median_heldout. Well below 1 suggests synthetic samples sit unusually close to training samples (possible memorization).

- too_close_fraction: Fraction of synthetic samples closer to a training sample than the 5th percentile of held-out DCRs. About 0.05 is expected with no copying; much higher suggests memorization. Evidence, not a privacy guarantee.

- mean_corr: Pearson correlation, across features, between per-feature means of synthetic and held-out real data (transformed space).

- var_corr: Same, for per-feature variances.

- zero_frac_corr: Same, for each feature's fraction of exact zeros (raw counts; only for models that generate counts).

- library_median_ratio: Median total count per synthetic sample divided by the held-out median (counts only).

- train_mean: Mean per-sample ELBO (VAE, a lower bound) or exact log-likelihood (flow) on training data, in the model's own input space. Not comparable across models with different input spaces.

- heldout_mean: Same quantity on held-out real data; a large drop from train_mean indicates overfitting.

- trustworthiness: How well the latent space avoids placing held-out samples near each other when they are not among each other's 15 nearest neighbors in the full transformed space (1 = perfectly, ~0.5 = random).

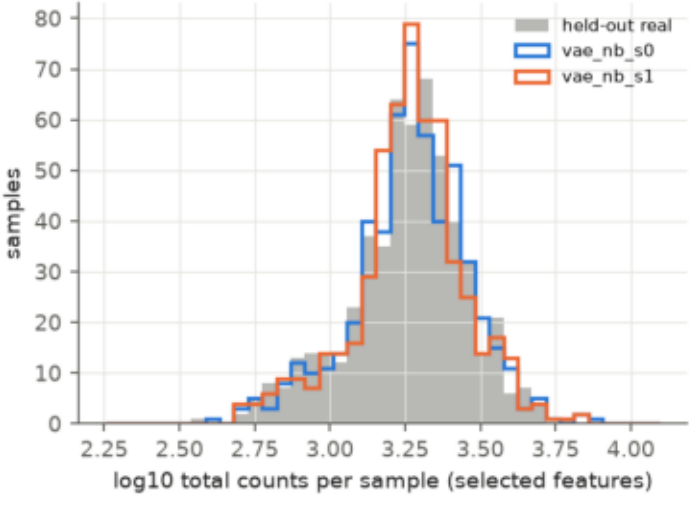
How to read the metrics (cont.)

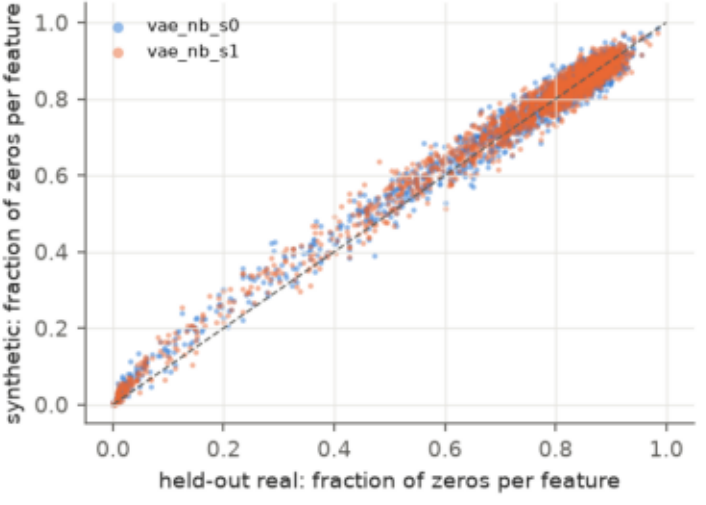
- trustworthiness_pca_baseline: Trustworthiness of a PCA (fit on training data) with as many dimensions as the latent space: the bar a learned representation is compared with.

- knn_overlap: Average fraction of each held-out sample's 15 nearest neighbors in the full transformed space that are also among its 15 nearest neighbors in the latent space.

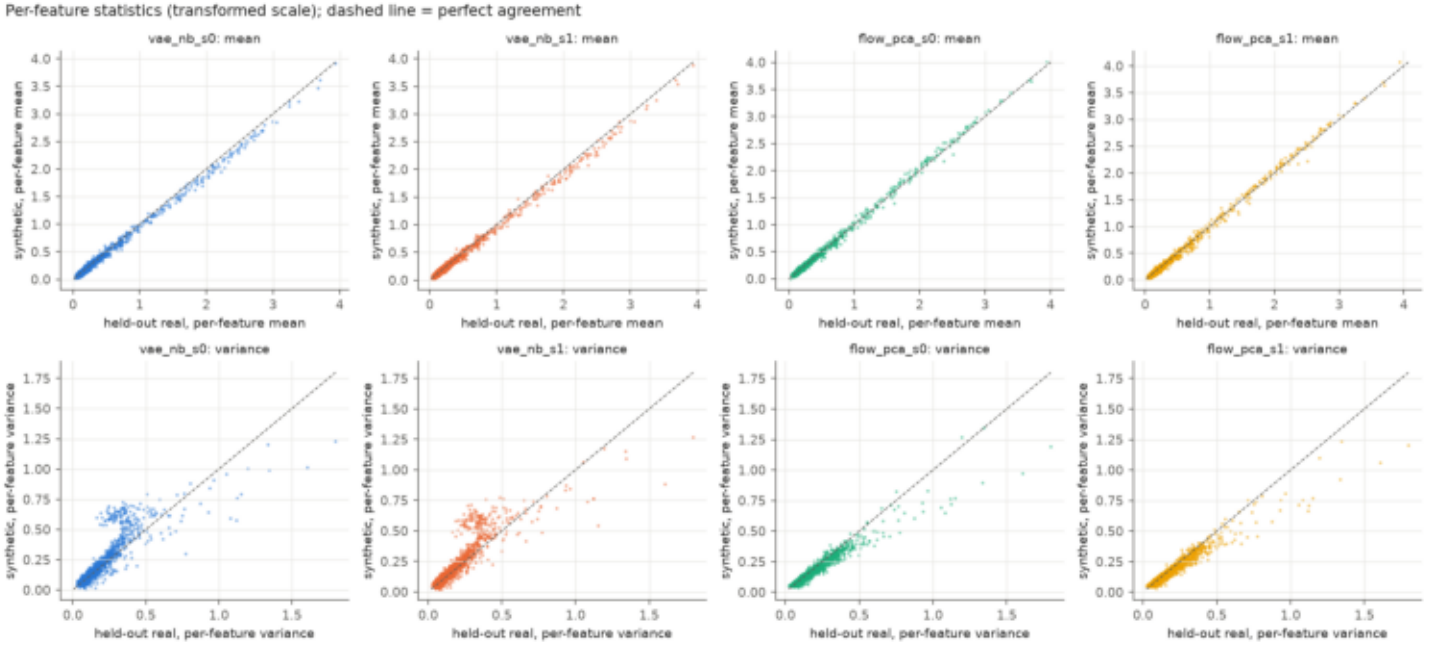
- knn_overlap_pca_baseline: Same, for the same-size PCA baseline.

Count-level properties

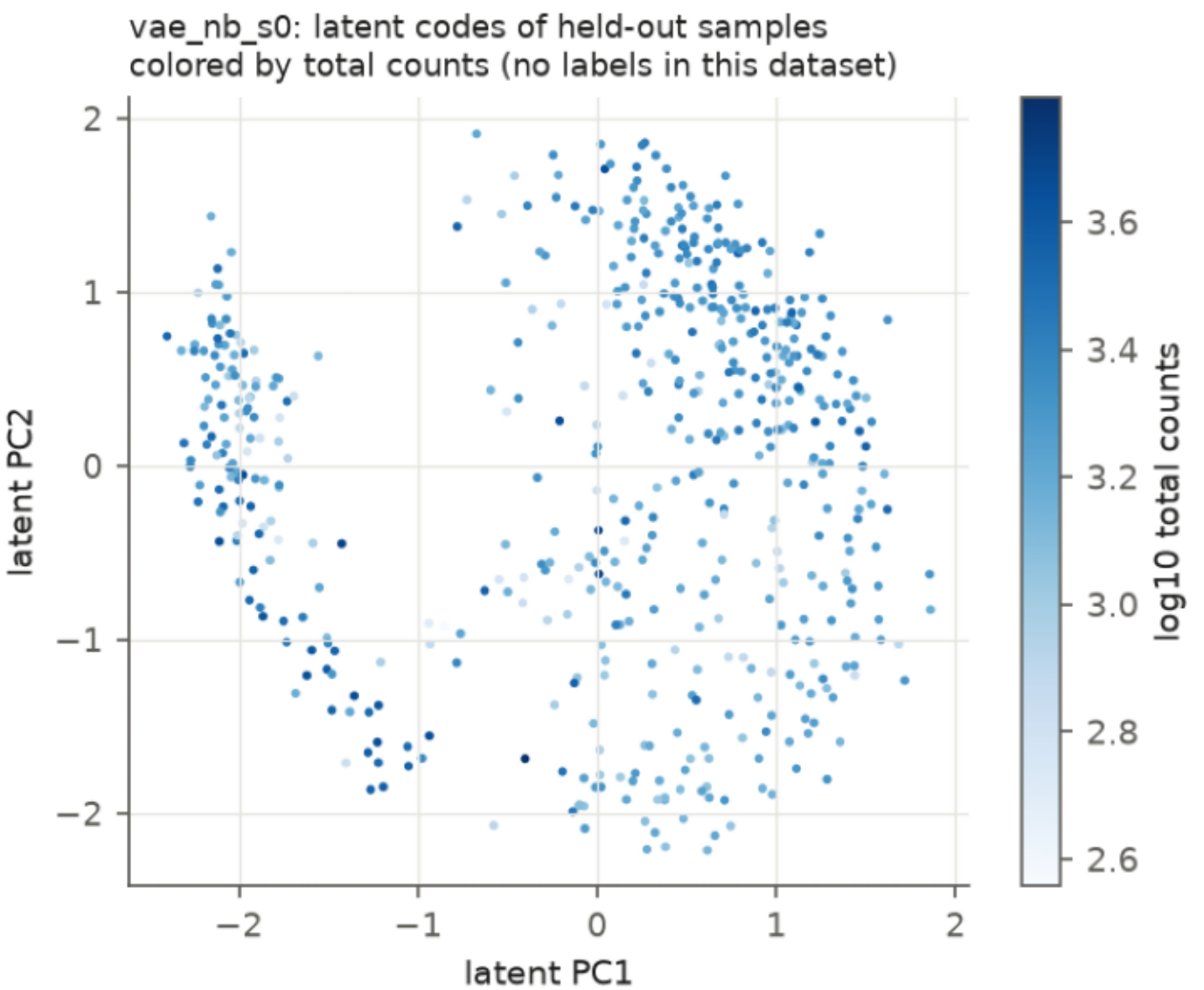




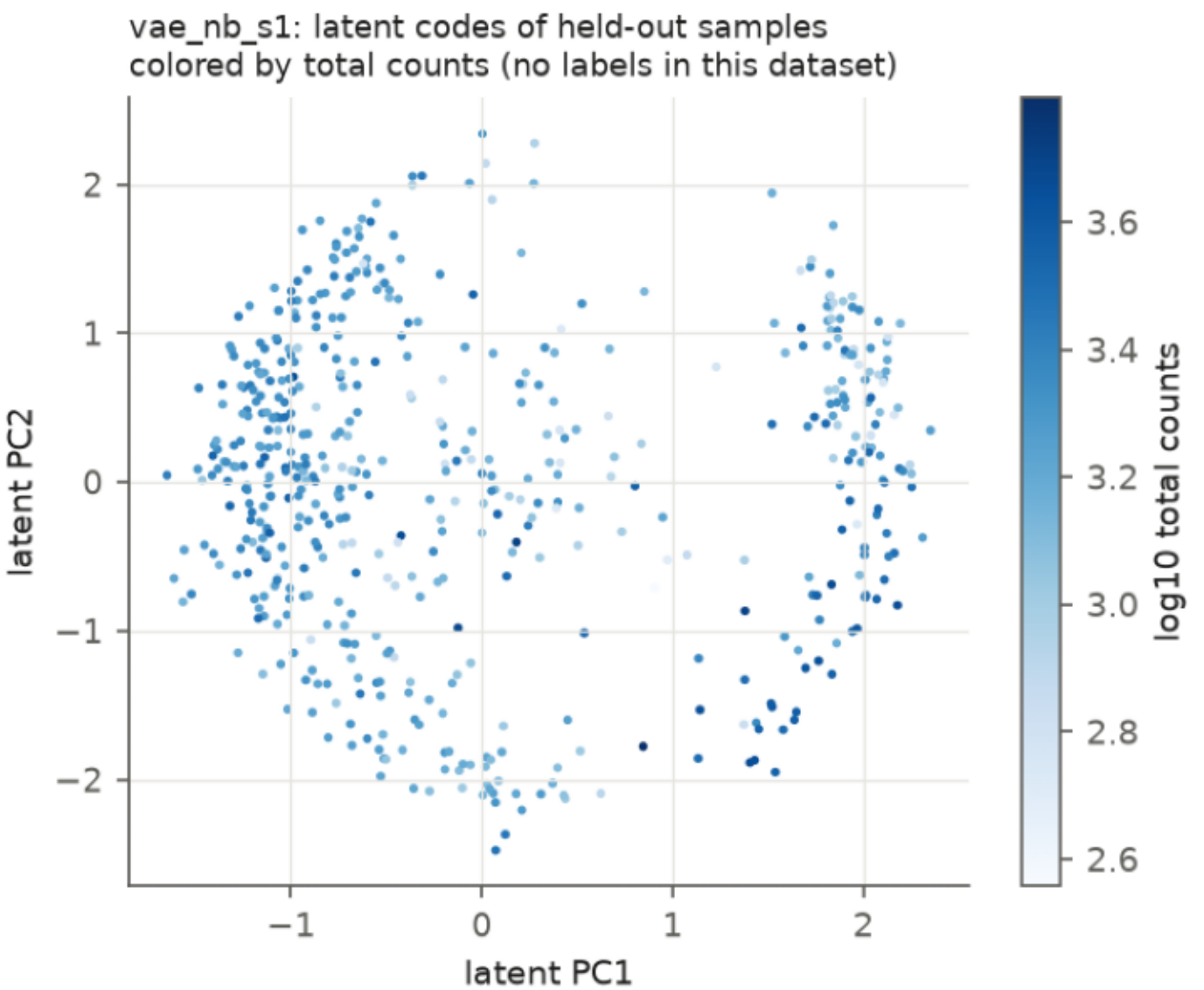
Count-level checks for models that generate counts: total counts per sample, and per-feature fraction of zeros.



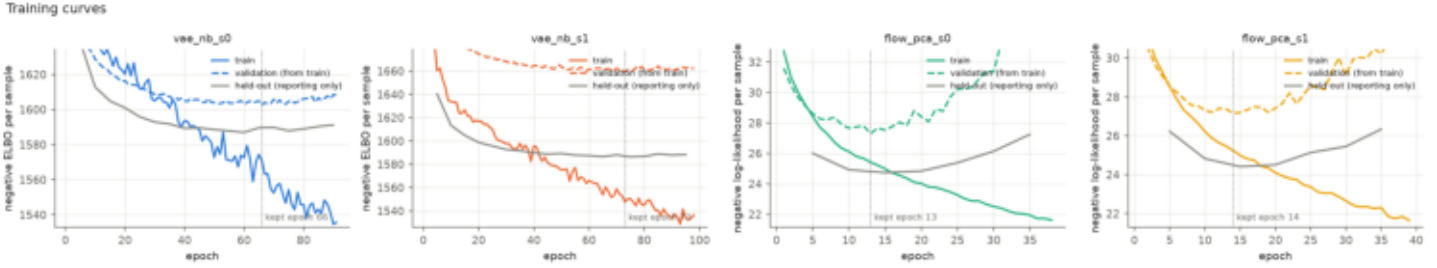
Per-feature mean and variance of synthetic vs held-out real data on the transformed scale. Points on the dashed line agree exactly.



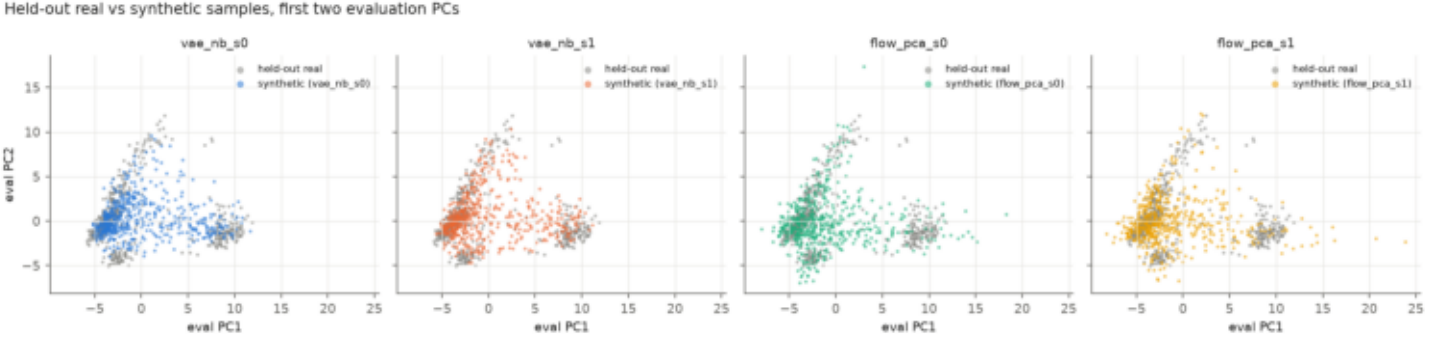
Two-dimensional PCA of a VAE's latent codes for held-out samples. A picture for orientation; neighbor-preservation numbers are on the metrics page.



Two-dimensional PCA of a VAE's latent codes for held-out samples. A picture for orientation; neighbor-preservation numbers are on the metrics page.



Training curves. The validation set is 10% of the training split and decides early stopping; the held-out curve is shown for reference only and never used to choose anything.



Held-out real samples (gray) and synthetic samples in the first two components of the shared evaluation space. Overlap in two components is necessary but not sufficient; the C2ST and MMD metrics use all components.

Findings

What each model produced

NB-VAE, seed 0 (vae_nb_s0): Stopped early at epoch 91 because validation loss had not improved for 25 epochs; the weights from epoch 66 were kept. Held-out ELBO -1586.174 against training -1563.406, a small gap. C2ST AUC 0.9457 (reference 0.5855); MMD2 0.01961 (reference 0.00026). Precision 0.7926, recall 0.7667. Memorization: DCR ratio 1.4897, too-close fraction 0.0056 (about 0.05 expected with no copying). Per-gene mean correlation 0.9978, variance correlation 0.8745, zero-fraction correlation 0.9919, median library size ratio 0.9984. Latent trustworthiness 0.8073 vs PCA baseline 0.8328; kNN overlap 0.16 vs 0.1681.

NB-VAE, seed 1 (vae_nb_s1): Stopped at epoch 98, kept epoch 73. Held-out ELBO -1585.719 against training -1561.089. C2ST AUC 0.9253 (reference 0.5664); MMD2 0.01639 (reference -0.00028). Precision 0.8722, recall 0.7667. DCR ratio 1.4131, too-close fraction 0.0019. Variance correlation 0.8912. Latent trustworthiness 0.7917 vs PCA baseline 0.8327; kNN overlap 0.1475 vs 0.1677.

Flow on PCA, seed 0 (flow_pca_s0): Stopped at epoch 38, kept epoch 13. The 20 input components keep 0.2396 of the transformed-data variance. C2ST AUC 0.8572 (reference 0.5855); MMD2 0.00767 (reference 0.00026). Precision 0.5611, recall 0.987. DCR ratio 1.5933, too-close fraction 0.0. Variance correlation 0.9694.

Flow on PCA, seed 1 (flow_pca_s1): Stopped at epoch 39, kept epoch 14. C2ST AUC 0.8352 (reference 0.5664); MMD2 0.00959 (reference -0.00028). Precision 0.5315, recall 0.9833. DCR ratio 1.5531, too-close fraction 0.0074. Variance correlation 0.9718.

All four runs finished well inside the 20-minute budget (1.12 minutes in total per run_log.json); none hit the time cap.

What the figures show

Real vs synthetic (first two evaluation PCs): held-out real cells form a few dense groups with empty space between them. VAE samples cover the main groups but also fill the space between them, so the groups are less distinct than in real data. Flow samples cover the real groups widely but scatter further out, with some points far beyond any real cell.

Training curves: both VAE seeds show validation loss flattening around epoch 40 to 70 while training loss keeps falling, and early stopping kept a point near the minimum. Both flows reach their best validation loss by epoch 13 to 14, then validation loss climbs steeply while training loss keeps falling, a clear sign of fast overfitting on 1,944 training cells; the held-out curve follows the same rise.

Per-feature statistics: per-gene means lie on the diagonal for all four runs. For variances, the flows track the diagonal closely, while both VAE seeds show a distinct group of genes with synthetic variance near 0.5 to 0.75 where the real variance is around 0.25 to 0.5.

Count properties (VAE only): the synthetic library-size histogram overlaps the held-out one closely, and per-gene zero fractions sit near the diagonal, slightly above it for genes with intermediate sparsity.

Latent space (VAE, colored by total counts since there are no labels): both seeds show one compact, separate group and a larger, more diffuse region. Total counts vary within both

Findings (cont.)

regions with no strong gradient. The layouts differ between seeds (mirrored and rotated), as expected for a VAE. These groups are exploratory structure only; no cell types can be assigned without labels.

How the models compare

Fidelity, memorization, and feature statistics are computed in the same 14-component evaluation space, so they can be compared. The flow is harder to tell from real cells than the VAE (C2ST AUC 0.8572 and 0.8352 vs 0.9457 and 0.9253) and has lower MMD and better variance agreement, but its samples are less often near a real cell (precision 0.5611 and 0.5315 vs 0.7926 and 0.8722), while it covers more of the real cells (recall 0.987 and 0.9833 vs 0.7667). In short, the VAE is more conservative but blurs the gaps between groups; the flow spreads wider and produces outliers. Neither comes close to the real-vs-real reference.

The VAE is the only model that produces integer counts and a latent representation, so it is the one that serves both parts of the objective. Likelihoods are not comparable across the two families: the VAE's ELBO is on 2,000 raw-count genes, and the flow's log-likelihood is on 20 PCA components.

Results are consistent across seeds for both families, so the conclusions do not depend on one lucky or unlucky run.

Answer to the objective

Partly. For the representation, the held-out ELBO stays close to the training value, so the model generalizes, but the latent space does not beat a same-size PCA on neighborhood preservation (0.8073 and 0.7917 vs about 0.833), so the success criterion is only partly met. For generation, the criterion of a C2ST AUC close to 0.5 is not met by any model (lowest 0.8352 vs reference 0.5664). The memorization part is met: synthetic cells are further from training cells than held-out real cells are (DCR ratio 1.4131 to 1.5933) and too-close fractions are well below 0.05. That is evidence against copying, not a privacy guarantee, and because fidelity failed, the synthetic cells are not recommended for sharing in place of the real data.

Downstream use of the generated data

VAE latent codes as input to clustering or neighbor graphs

Verdict: CONDITIONAL | task: representation_learning | model: vae_nb_s0

The latent space preserves held-out neighborhoods slightly worse than a 10-component PCA of the log-normalized data in both seeds, so for neighbor-based uses a PCA would do about as well. The held-out ELBO is close to the training value (a gap of about 23 nats per cell out of about 1,586), so the model generalizes rather than memorizing; its added value is a count-aware model of raw UMIs that also comes with a decoder.

- evidence: vae_nb_s0.representation.trustworthiness = 0.8073 (vs vae_nb_s0.representation.trustworthiness_pca_baseline = 0.8328)

- evidence: vae_nb_s0.representation.knn_overlap = 0.16 (vs vae_nb_s0.representation.knn_overlap_pca_baseline = 0.1681)

- evidence: vae_nb_s1.representation.trustworthiness = 0.7917 (vs vae_nb_s1.representation.trustworthiness_pca_baseline = 0.8327)

- evidence: vae_nb_s0.likelihood.heldout_mean = -1586.174 (vs vae_nb_s0.likelihood.train_mean = -1563.406)

- caveat: Only use it if a count-likelihood representation (library size handled in the decoder) is specifically wanted; otherwise PCA on log-normalized counts is the simpler, slightly better choice for Euclidean neighbor structure.

- caveat: Trustworthiness is measured in Euclidean log-normalized space, which favors PCA by construction; the VAE could still preserve biologically meaningful groups that this metric does not reward, but no labels exist to check that.

- caveat: The two seeds give different latent layouts (see latent figures); downstream clusters should be checked for stability across seeds.

- caveat: No QC was applied to the raw cells, so low-quality cells or doublets are embedded along with everything else.

VAE synthetic cells in place of real cells (incl. sharing)

Verdict: NOT_RECOMMENDED | task: data_generation | model: vae_nb_s0

A classifier separates synthetic from held-out real cells almost perfectly in both seeds, far above the real-vs-real reference, and MMD is far above its reference. The scatter of the first two evaluation PCs shows synthetic cells filling the gaps between real cell populations, so cluster structure and proportions in the synthetic data would not match the real data.

- evidence: vae_nb_s0.fidelity.c2st_auc = 0.9457 (vs vae_nb_s0.fidelity.c2st_auc_reference = 0.5855)

- evidence: vae_nb_s1.fidelity.c2st_auc = 0.9253 (vs vae_nb_s1.fidelity.c2st_auc_reference = 0.5664)

- evidence: vae_nb_s0.fidelity.mmd2 = 0.01961 (vs vae_nb_s0.fidelity.mmd2_reference =

Downstream use of the generated data (cont.)

0.00026)

- caveat: This verdict holds for both VAE seeds.

- caveat: Biological realism (cell-type composition, marker co-expression) could not be checked directly because the dataset has no cell-type labels.

VAE synthetic cells for per-gene count summaries only

Verdict: CONDITIONAL | task: data_generation | model: vae_nb_s0

Per-gene means, per-gene zero fractions, and the median library size of synthetic cells closely match held-out real cells, and the output is integer counts, so the samples reproduce single-gene count properties. Per-gene variances agree less well: the per-feature statistics figure shows a group of genes whose synthetic variance is roughly double the real variance.

- evidence: vae_nb_s0.feature_stats.mean_corr = 0.9978 (vs perfect agreement = 1)

- evidence: vae_nb_s0.feature_stats.zero_frac_corr = 0.9919 (vs perfect agreement = 1)

- evidence: vae_nb_s0.feature_stats.library_median_ratio = 0.9984 (vs perfect agreement = 1)

- evidence: vae_nb_s0.feature_stats.var_corr = 0.8745 (vs flow_pca_s0.feature_stats.var_corr = 0.9694)

- caveat: Only for uses that look at one gene at a time; any analysis that combines genes falls under the not_recommended use above.

- caveat: Per-gene variance is inflated for a subset of genes, so variance-based steps (e.g. highly variable gene selection) on synthetic cells may pick different genes than on real cells.

- caveat: This is not a basis for sharing in place of real data; see the privacy statement.

Flow synthetic cells in place of real cells

Verdict: NOT_RECOMMENDED | task: data_generation | model: flow_pca_s1

The flow is somewhat harder to separate from real cells than the VAE (lower C2ST AUC, better variance agreement) but is still far above the real-vs-real reference, and only about half of its samples fall near a real cell, consistent with the far outliers visible in the evaluation-PC scatter. It produces continuous log-scale values, not counts, so it cannot serve as synthetic raw count data.

- evidence: flow_pca_s1.fidelity.c2st_auc = 0.8352 (vs flow_pca_s1.fidelity.c2st_auc_reference = 0.5664)

- evidence: flow_pca_s1.fidelity.precision = 0.5315 (vs vae_nb_s1.fidelity.precision = 0.8722)

- evidence: flow_pca_s0.fidelity.c2st_auc = 0.8572 (vs flow_pca_s0.fidelity.c2st_auc_reference = 0.5855)

- caveat: The flow models only the top 20 PCA components (about 24% of transformed-data

Downstream use of the generated data (cont.)

variance per the run sidecar) and adds Gaussian noise for the rest, so gene-level structure outside those components is not learned.

- caveat: Validation loss rose quickly after epoch 13 to 14 in both seeds; early stopping kept the best weights, but the model overfits the 1,944 training cells fast.

Not tested (no recommendation either way)

- Whether clusters or cell-type proportions found in synthetic cells match those in real cells: would need cell-type labels or a reference annotation, which this dataset lacks.

- Whether a downstream model trained on synthetic cells performs as well on real cells (train-on-synthetic, test-on-real): there is no labeled downstream task here.

- Differential expression or marker-gene discovery on synthetic cells: not evaluated; given the C2ST results, gene-gene structure is not reliable.

- Any formal privacy property (e.g. differential privacy, membership-inference resistance): not tested; only nearest-neighbor distances were measured.

- Robustness to the missing QC step: low-quality cells and doublets in the raw data were modeled as-is.

Privacy

There is no formal privacy guarantee for any output of this run: the models were not trained with differential privacy or any other formal method, and no membership-inference test was run. The pipeline only checked memorization by distance in the evaluation space. Synthetic VAE cells are not unusually close to training cells: the median distance from a synthetic cell to its nearest training cell is 1.49 times (seed 0) and 1.41 times (seed 1) the distance for held-out real cells, and the fraction of synthetic cells closer to a training cell than the 5th percentile of held-out distances is 0.0056 and 0.0019, against about 0.05 expected with no copying. This is evidence that the models did not copy training cells, not proof that information about individual donors or cells cannot be recovered. Any release of synthetic cells derived from this data would need a separate privacy review or a formally private training method, and given the fidelity results above the synthetic cells are not a substitute for the real data in any case.

Limitations

- Fidelity is measured in a 14-component PCA space (0.2258 of training variance); differences outside it are not measured.
- No independent critique was run for this report.
- The raw data has no quality control: low-quality cells and possible doublets were modeled as-is, and some of the structure the models learned may reflect them.
- There are no labels, so biological realism (cell-type proportions, marker co-expression) could not be tested, and latent groups cannot be named.
- The flow models only the top 20 components and adds independent Gaussian noise for the rest, and it outputs continuous values, not counts.
- No formal privacy method was used. The memorization check is a distance comparison and does not rule out other kinds of information leakage.

Reproducibility

Command: ./run.sh pbmc "Learn a low-dimensional representation and generate realistic synthetic cells for sharing" --report-number 1

Plan seed 0; plan revision 1; budget 20 min; pipeline time 1.12 min

Preprocessing hash 93b023f05bc08e80

python 3.13.5, torch 2.14.0, numpy 2.5.3, scikit-learn 1.9.1

- vae_nb_s0: seed 0, config hash 1e3e55c97da71f7c, 0.194 min on cpu

- vae_nb_s1: seed 1, config hash 4184940d0ff3beed, 0.194 min on cpu

- flow_pca_s0: seed 0, config hash 95a7adb96933bcfd, 0.023 min on cpu

- flow_pca_s1: seed 1, config hash cee5bfb809e3621c, 0.024 min on cpu

Generated 2026-10-23 15:44. All intermediate files: outputs/pbmc/.